

P2 - A Decision Analysis Approach for Optimizing School Garden Interventions - Addressing Child Health, Biodiversity, and Economic Outcomes

Cory Whitney, Johannes Kopton, Eike Luedeling, Thi Thu Giang Luu

Institute of Crop Science and Resource Conservation, Horticultural Sciences (INRES), University of Bonn

This study evaluates the impacts of school gardens in Vietnam, focusing on child health, biodiversity, and economic outcomes across public and private schools. We assessed five investment scenarios using decision analysis, including school gardens with and without integration of STEM (Science, Technology, Engineering, and Math) education into the garden. It includes specialized teaching equipment and teacher training to enhance learning experiences. curriculum and schools without gardens. We mapped impact pathways to identify trade-offs, risks, and uncertainties, employing Monte Carlo simulations, sensitivity analysis, Pareto optimization, and Expected Value of Perfect Information (EVPI) to guide decision-making.

Our Pareto optimization model revealed trade-offs among economic, biodiversity, and health outcomes. Gardens without STEM integration consistently provided greater health benefits. STEM gardens, particularly in public schools, faced financial and operational challenges. Key success drivers included garden-related events, teacher training, and community support, highlighting the critical role of targeted investments in addressing uncertainties in economic returns. We used Non-dominated Sorting Genetic Algorithm II (NSGA-II) to explore 2500 Monte Carlo simulation runs to approximate Pareto fronts for each scenario. Optimal configurations were reached when garden size and frequency of public school events were maximized. Integrating animals in the garden increased child health benefits while reducing overall economic return. Public schools showed strong potential for biodiversity improvements with minimal economic trade-offs, while private schools demonstrated higher economic feasibility, especially when leveraging gardens to enhance reputation and enrollment. Our integrated analytical approach provides actionable insights for policymakers and educators to optimize school garden interventions. These approaches can foster sustainable food environments and improve child health.

Contact:

Dr. Cory Whitney

email: cory.whitney@uni-bonn.de